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# Tau Ceti grain-size physics and collisional audit

## Executive result

The 20 um value is not a physically derived Tau Ceti radiation-pressure blowout diameter. Source inventory
The project's accepted f/f_max reconciliation remains untouched. This audit is isolated and uses the same repro

## Physical blowout calculation

For diameter D,  $\beta_{\text{rad}} = 3 L Q_{\text{pr}} / (8 \pi c G M \rho D)$ , so  $\beta=0.5$  gives  $D_{\text{bl}} = 3 L Q_{\text{pr}} / (4 \pi c G M \rho)$ . The
Adding a solar-wind pressure term with the documented solar reference and a Tau-specific upper limit of 0.1 sol

## D_bl, D_min and D_char

- D_bl: radiation-pressure blowout diameter at  $\beta=0.5$ , derived from L, M, density, Q_pr and optional wind
- D_min: lower cutoff of the collisional size distribution. Lawler et al. report 15 +/- 8 um from a physical
- D_char: characteristic emitting size; not constrained by the current modified-blackbody SED. Temperature,

For  $q=11/6$ , the implemented Eq. 14 has f_max proportional to  $D_{\text{cut}}^{-0.5}$ , so f/f_max is proportional to  $D_{\text{cut}}^{+0.5}$ 

## Controlled collisional matrix

The matrix in `tables/controlled_model_matrix.csv` separates luminosity, geometry, physical D_bl/D_min and diag
The D_min grid, reference and controlled diagnostic-luminosity sensitivity are in `tables/dmin_sensitivity_ring

## Final recommendation

Use the physical D_bl calculation (with explicit density/Q_pr scenarios and the wind bound) as the primary

## Reproducibility and limitations

Run from the project environment with:

```bash
/Users/romonedunlop/Documents/Tau\ Ceti\ Research/.venv/bin/python /Users/romonedunlop/Documents/Tau\ Ceti\ Res
```

The physical grid is analytic rather than Mie-based; Teff and stellar radius are not used because they are not

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